KAKSHA PRACTICE PAPER · CLASS 10

Science Mock Test 1

Time allowed: 3 hours **Maximum marks: 80**

Five sections A–E. Section A: 20 one-mark questions (16 MCQs and 4 assertion–reason). Section B: six 2–mark very short answer questions. Section C: seven 3–mark short answer questions. Section D: three 5–mark long answer questions. Section E: three case-based questions of 4 marks.

Section A

Multiple-choice questions (Q1–16) and assertion–reason questions (Q17–20). Choose the correct option.

1. The reaction $Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$ is an example of **[1]**

(a) a combination reaction
(b) a decomposition reaction
(c) a displacement reaction
(d) a neutralisation reaction

2. When zinc granules react with dilute hydrochloric acid, the gas evolved is **[1]**

(a) hydrogen
(b) oxygen
(c) chlorine
(d) carbon dioxide

3. A solution turns red litmus blue. Its pH is most likely **[1]**

(a) 1
(b) 4
(c) 7
(d) 10

4. The chemical formula of baking soda is **[1]**

(a) Na_2CO_3
(b) $NaHCO_3$
(c) $Na_2CO_3 \cdot 10H_2O$
(d) NaOH

5. Which metal is stored under kerosene to prevent it from catching fire in air? **[1]**

(a) Copper
(b) Iron
(c) Sodium
(d) Aluminium

6. The functional group present in ethanoic acid is **[1]**

(a) –OH
(b) –CHO
(c) C=O (ketone)
(d) –COOH

7. The number of covalent bonds in a molecule of ethane (C_2H_6) is **[1]**

(a) 6
(b) 7
(c) 8
(d) 9

8. In human lungs, the exchange of gases takes place in the **[1]**

(a) bronchi
(b) trachea
(c) alveoli
(d) larynx

9. The plant hormone that promotes cell division is **[1]**

(a) cytokinin
(b) auxin
(c) abscisic acid
(d) ethylene

10. The small gap between the nerve ending of one neuron and the dendrite of the next is called a **[1]**

(a) axon
(b) cell body
(c) synapse
(d) nerve fibre

11. Buds produced in the notches along the leaf margin, which develop into new plants, are found in **[1]**

(a) potato
(b) ginger
(c) rose
(d) Bryophyllum

12. In Mendel's dihybrid cross between round-yellow and wrinkled-green pea plants, the F_2 phenotypic ratio is **[1]**

(a) 3 : 1
(b) 1 : 2 : 1
(c) 9 : 3 : 3 : 1
(d) 1 : 1 : 1 : 1

13. A concave mirror forms a real, inverted image of the same size as the object when the object is placed **[1]**

(a) at the focus
(b) between the focus and the pole
(c) at the centre of curvature
(d) beyond the centre of curvature

14. The power of a concave lens of focal length 50 cm is **[1]**

(a) +2 D
(b) −2 D
(c) −0.5 D
(d) +0.5 D

15. The rule used to find the direction of force on a current-carrying conductor placed in a magnetic field is **[1]**

(a) right-hand thumb rule
(b) Fleming's right-hand rule
(c) Fleming's left-hand rule
(d) Snell's law

16. Which of the following is biodegradable? **[1]**

(a) Polythene bag
(b) Aluminium foil
(c) Glass bottle
(d) Vegetable peels

17. Assertion (A): Pepsin can digest proteins in the stomach. Reason (R): Hydrochloric acid in gastric juice creates the acidic medium needed for pepsin to act. **[1]**

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

18. Assertion (A): The chemical formula of Plaster of Paris is $CaSO_4 \cdot 2H_2O$. Reason (R): Plaster of Paris sets into a hard mass when mixed with water. **[1]**

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

19. Assertion (A): In a series circuit, the same current flows through every resistor. Reason (R): The equivalent resistance of resistors in series is less than the smallest individual resistance. **[1]**

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

20. Assertion (A): The human heart has four chambers. Reason (R): The wall of the left ventricle is thicker than that of the right ventricle. **[1]**

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Section B

Very short answer questions.

21. Write the balanced chemical equation for the reaction of quicklime with water. Name the type of reaction and state whether heat is absorbed or released. **[2]**

22. Why should curd and other sour substances not be kept in brass or copper vessels? **[2]**

23. Give any two differences between arteries and veins. **[2]**

24. State the role of (i) bile juice and (ii) villi in human digestion. **[2]**

25. The refractive index of glass is 1.5. Find the speed of light in glass. (Speed of light in vacuum = 3×10^8 m/s) **[2]**

26. Why is the ozone layer important? Name the chemicals mainly responsible for its depletion. **[2]**

Section C

Short answer questions.

27. What are amphoteric oxides? Give one example and write equations showing its reaction with an acid and a base. **[3]**

28. What is esterification? Write the chemical equation for the reaction of ethanol with ethanoic acid. What happens when the product is treated with sodium hydroxide? **[3]**

29. What is meant by double circulation? Why is it necessary in human beings? **[3]**

30. Mendel crossed pure tall pea plants (TT) with pure short pea plants (tt). What did he observe in the F_1 and F_2 generations? What conclusion did he draw? **[3]**

31. An object is placed 20 cm in front of a convex lens of focal length 15 cm. Find the position and nature of the image and its magnification. **[3]**

32. Three resistors of 2 Ω, 3 Ω and 6 Ω are connected in parallel to a 6 V battery. Find (i) the equivalent resistance (ii) the total current (iii) the current through the 2 Ω resistor. **[3]**

33. List any three properties of magnetic field lines. **[3]**

Section D

Long answer questions.

34. What is the chlor-alkali process? Write the balanced equation, name the three products, state where each is formed and give one use of each. **[5]**

35. Describe the process of fertilisation in flowering plants. What changes take place in the flower after fertilisation? **[5]**

36. What is myopia? State two causes. How is it corrected? A myopic person cannot see objects clearly beyond 80 cm. Find the nature and power of the lens needed. **[5]**

Section E

Case-study questions (1 + 1 + 2 marks).

37. Case study — The baker's secret. A baker adds a white powder to cake batter. When the cake is baked, it becomes soft and spongy. The same powder is also used in antacids. When this powder is heated strongly, it gives a substance used to make washing soda. (i) Give the chemical name and formula of the powder. (1) (ii) Why does the cake become soft and spongy? (1) (iii) Write the equation for the effect of heat on the powder, and how washing soda is obtained from the product. (2) **[4]**

38. Case study — Chemical messengers. A doctor tells a patient with a swollen neck to use iodised salt. Another patient has high blood sugar and is advised insulin injections. A third person notices that when frightened, his heart beats fast and he breathes faster. (i) Why is iodised salt advised for the first patient? (1) (ii) Which gland secretes insulin and which disease results from its deficiency? (1) (iii) Name the hormone responsible for the third person's response, its gland, and two of its effects. (2) **[4]**

39. Case study — Electricity bill. A house uses four bulbs each rated 100 W, 220 V. Each bulb is used for 5 hours daily. (i) Find the current drawn by one bulb. (1) (ii) Find the resistance of the filament of one bulb. (1) (iii) Find the electrical energy consumed by the four bulbs in 30 days, and its cost at ₹5 per unit. (2) **[4]**

Answer key and marking points

1. (c) a displacement reaction

2. (a) hydrogen

3. (d) 10

4. (b) $NaHCO_3$

5. (c) Sodium

6. (d) –COOH

7. (b) 7

8. (c) alveoli

9. (a) cytokinin

10. (c) synapse

11. (d) Bryophyllum

12. (c) 9 : 3 : 3 : 1

13. (c) at the centre of curvature

14. (b) –2 D

15. (c) Fleming's left-hand rule

16. (d) Vegetable peels

17. (a) Both A and R are true, and R is the correct explanation of A

18. (d) A is false but R is true

19. (c) A is true but R is false

20. (b) Both A and R are true, but R is not the correct explanation of A

21. $CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + heat$. (1)
It is a combination reaction, and it is exothermic (heat is released). (1)

22. Curd and sour substances contain acids. (½)
These acids react with copper/brass (zinc) to form poisonous, harmful salts that spoil the food and can make a person ill. (1½)

23. Arteries carry blood away from the heart; veins carry blood to the heart. (1)
Arteries have thick, elastic walls because blood is under high pressure; veins have thin walls and valves to prevent backflow. (1)

24. (i) Bile (from the liver) makes the food alkaline so pancreatic enzymes can act, and emulsifies fats into small globules. (1)
(ii) Villi are finger-like projections in the small intestine that increase the surface area for absorption of digested food. (1)

25. $n = c/v$, so $v = c/n$. (1)
$v = (3 \times 10^8)/1.5 = 2 \times 10^8$ m/s. (1)

26. The ozone layer in the upper atmosphere absorbs most of the harmful ultraviolet (UV) radiation from the Sun, which can cause skin cancer and harm living beings. (1)
Chlorofluorocarbons (CFCs), used in refrigerants and fire extinguishers, are mainly responsible for its depletion. (1)

27. Metal oxides that react with both acids and bases to give salt and water are amphoteric oxides. (1)
Example: aluminium oxide. $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$. (1)
$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$ (sodium aluminate). (1)

28. The reaction of a carboxylic acid with an alcohol in the presence of an acid catalyst to form a sweet-smelling ester is esterification. (1)
$CH_3COOH + CH_3CH_2OH \rightarrow CH_3COOCH_2CH_3 + H_2O$ (in presence of conc. H_2SO_4). (1)
With NaOH, the ester is converted back to ethanol and sodium ethanoate: $CH_3COOC_2H_5 + NaOH \rightarrow C_2H_5OH + CH_3COONa$. This is saponification. (1)

29. Blood passes through the heart twice in one complete cycle of the body – once through the pulmonary circuit (heart → lungs → heart) and once through the systemic circuit (heart → body → heart). (1½)
It keeps oxygenated and deoxygenated blood separate, giving a highly efficient supply of oxygen to the body. (1)
This is needed because birds and mammals are warm-blooded and need a lot of energy to maintain body temperature. (½)

30. F_1: all plants were tall (Tt). (1)
F_2 (selfing F_1): tall and short plants in the ratio 3 : 1; genotypes TT : Tt : tt = 1 : 2 : 1. (1)
Conclusion: traits are controlled by factors (genes) present in pairs; tallness is dominant and shortness is recessive; the recessive trait reappears in F_2, so factors are not lost. (1)

31. u = −20 cm, f = +15 cm. Lens formula: $1/v - 1/u = 1/f \Rightarrow 1/v = 1/15 - 1/20 = 1/60$. (1)
v = +60 cm, so the image is formed 60 cm on the other side of the lens. (1)
m = v/u = 60/(−20) = −3. The image is real, inverted and three times magnified. (1)

32. (i) $1/Rp = 1/2 + 1/3 + 1/6 = (3 + 2 + 1)/6 = 1$, so Rp = 1 Ω. (1)
(ii) $I = V/R = 6/1 = 6$ A. (1)
(iii) Voltage across each resistor is 6 V, so $I_2\Omega = 6/2 = 3$ A. (1)

33. Any three, 1 mark each: They emerge from the north pole and merge at the south pole outside the magnet (and go from south to north inside, forming closed loops). No two field lines ever intersect. The field is stronger where lines are closer together. The tangent at any point gives the direction of the field.

34. When electricity is passed through brine (aqueous sodium chloride), it decomposes to form sodium hydroxide; the process is called the chlor-alkali process (chlor for chlorine, alkali for NaOH). (1)
$2NaCl(aq) + 2H_2O(l) \rightarrow 2NaOH(aq) + Cl_2(g) + H_2(g)$. (1)
Chlorine gas is given off at the anode; hydrogen gas at the cathode; sodium hydroxide solution forms near the cathode. (1)
Uses: H_2 – fuels, margarine, ammonia for fertilisers; Cl_2 – water treatment, PVC, disinfectants, CFCs; NaOH – soaps and detergents, paper making, degreasing metals. (2)

35. Pollination: pollen grains are transferred from the anther to the stigma (self or cross pollination by wind, water, insects). (1)
A pollen tube grows out of the pollen grain, travels through the style and reaches the ovule in the ovary. (1)
The male germ cell fuses with the female germ cell (egg) in the ovule to form a zygote – this is

fertilisation. (1)
After fertilisation: the zygote divides to form an embryo; the ovule develops a tough coat and becomes the seed. (1)
The ovary grows rapidly and ripens to form the fruit; petals, sepals, stamens, style and stigma may shrivel and fall off. (1)

36. Myopia (near-sightedness): a person can see nearby objects clearly but not distant objects; the image of a distant object forms in front of the retina. (1)
Causes: excessive curvature of the eye lens; elongation of the eyeball. (1)
Correction: a concave (diverging) lens of suitable power, which brings the image back onto the retina. (1)
The lens must form the image of an object at infinity at the far point, so f = −80 cm = −0.8 m. (1)
P = 1/f = 1/(−0.8) = −1.25 D; a concave lens of power −1.25 D. (1)

37. (i) Sodium hydrogencarbonate, $NaHCO_3$ (baking soda). (1)
(ii) On heating, it releases carbon dioxide gas which gets trapped in the batter and makes it rise. (1)
(iii) $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$ (on heating). (1)
Recrystallising sodium carbonate gives washing soda: $Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3 \cdot 10H_2O$. (1)

38. (i) Iodine is needed by the thyroid gland to make thyroxine; its deficiency causes goitre (swollen neck). (1)
(ii) The pancreas secretes insulin; its deficiency causes diabetes. (1)
(iii) Adrenaline, secreted by the adrenal glands. (1)
Effects (any two): faster heartbeat to supply more oxygen to muscles; increased breathing rate; blood diverted to skeletal muscles; preparing the body for "fight or flight". (1)

39. (i) $I = P/V = 100/220 \approx 0.45$ A. (1)
(ii) $R = V^2/P = (220 \times 220)/100 = 484\ \Omega$. (1)
(iii) Energy = 4 × 100 W × 5 h × 30 = 60,000 Wh = 60 kWh = 60 units. (1)
Cost = 60 × ₹5 = ₹300. (1)